

Observability Lab: High-Cardinality Telemetry Cost Optimization & Sampling

Duration : 30 Min.

Scenario

Cloud observability costs have exceeded budget due to unindexed high-cardinality log labels and un-sampled trace ingestion.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Identify high-cardinality labels, configure head/tail trace sampling, and establish log-to-metric conversion rules to optimize costs.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the "**Submit Final**" button. This task is evaluated manually by the grading team.

Evaluation Rubric:

1. Identifying high-cardinality metric labels causing storage explosion: 3 marks
2. Configuring head-based and tail-based trace sampling strategies: 3 marks
3. Implementing log-to-metric conversions to drop verbose raw log lines: 2 marks
4. Setting log retention policies based on environment criticalities: 2 marks

Total Score: 10 Marks

Important Notes

- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.